

Language Translation System

English to Hindi

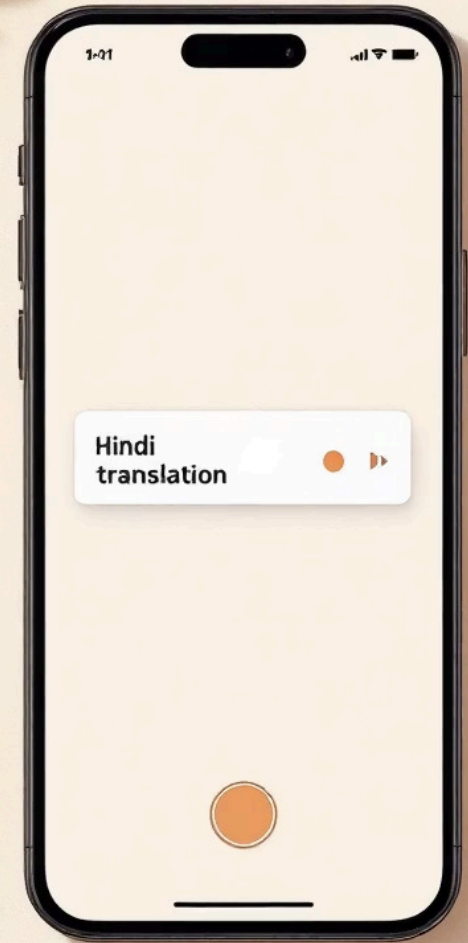
A transformer-based NLP system that enables seamless English-to-Hindi translation with near real-time performance.

🔧 NLP | TRANSFORMERS | FLASK

INVERTIS UNIVERSITY

BTECH CSE | SECTION C | GROUP J

TEAM LEAD | FATIMA SEEMAB



Made with GAMMA

The Language Barrier Problem

Communication Gap

Language differences create significant obstacles in information exchange, education, and global collaboration.

Manual Translation

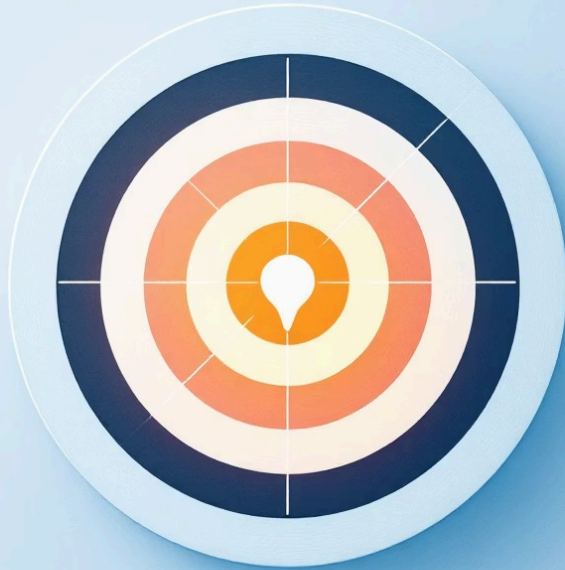
Human translation is slow, costly, and prone to inconsistencies, making it unsuitable for real-time communication.

The Need

A system that provides fast, accurate, and real-time translation is essential to bridge this gap.



Project Objectives



Accurate Translation

Develop an AI system that converts English text into Hindi with high precision using transformer-based models.

Real-Time Performance

Enable real-time, on-the-fly translation with minimal response delay through an interactive web interface

Accessible Interface

Build a clean, user-friendly interface accessible to non-technical people.

Technology Stack



Python & Flask

Backend logic and server-side handling of user requests and model inference



Hugging Face Transformers

Pre-trained NLP models for state-of-the-art sequence-to-sequence translation.



PyTorch

Underlying deep learning framework used by Transformers for model execution and tensor computations.



HTML & CSS

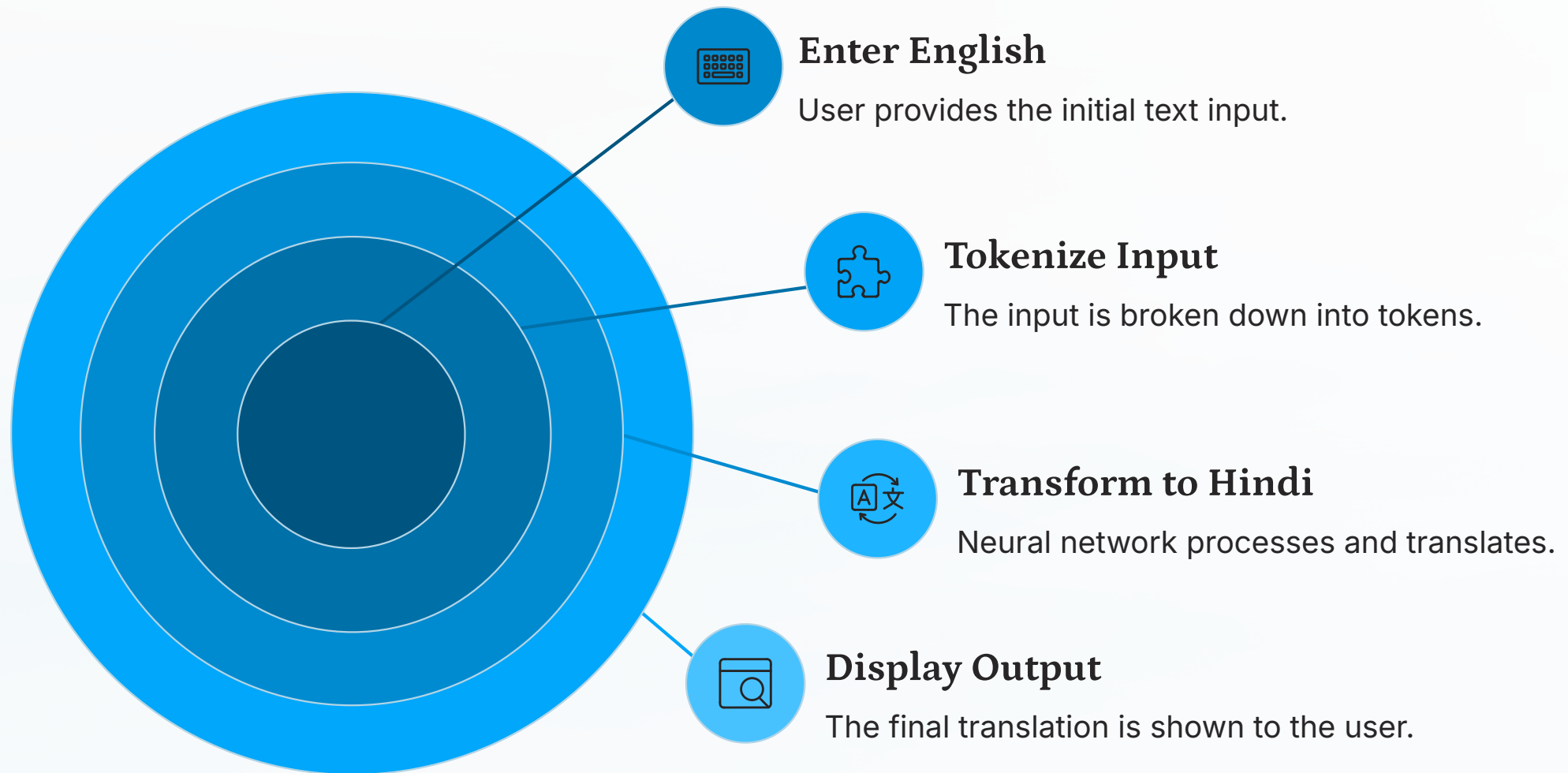
Lightweight frontend interface for user input and translation output display.



Hugging Face Spaces

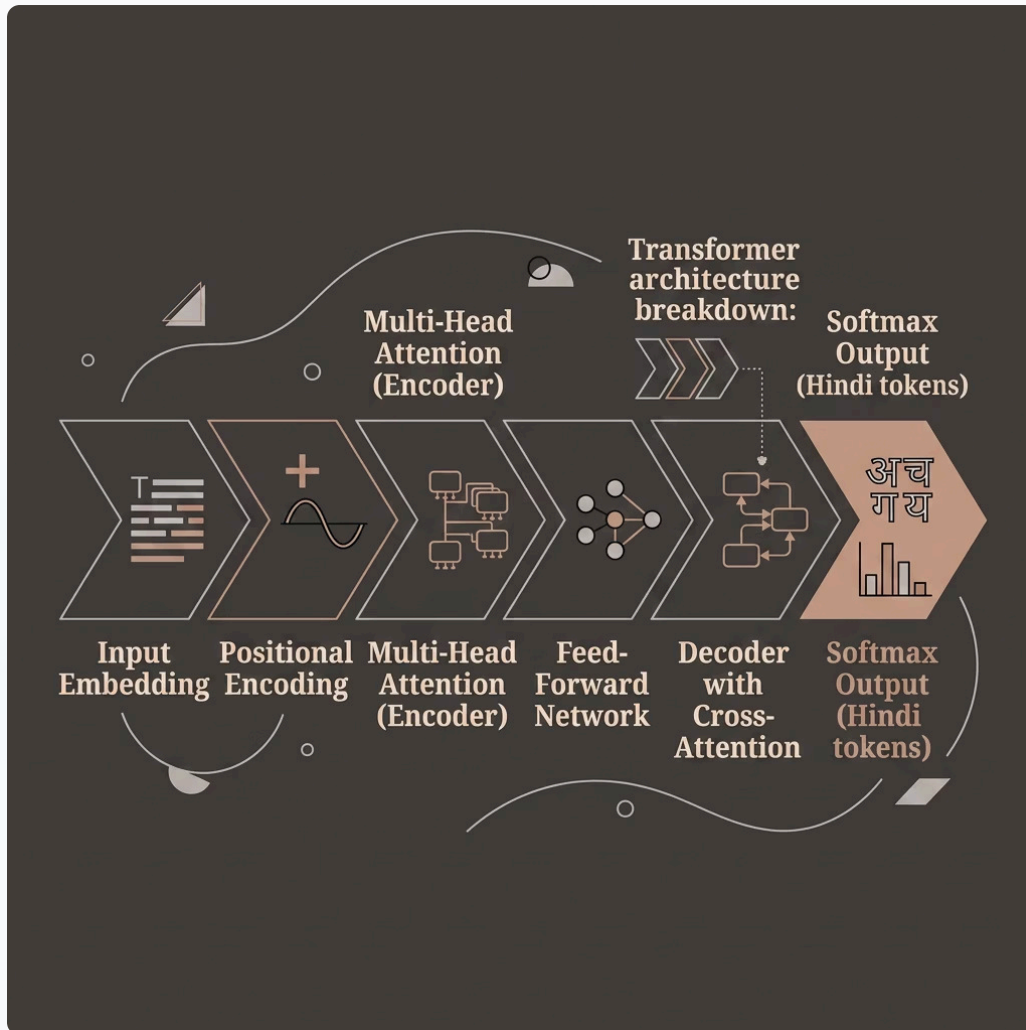
Cloud deployment platform for hosting and sharing the live application.

How It Works



The system uses a pre-trained **Seq2Seq** transformer model (MarianMT). Input English text is tokenized, passed through encoder-decoder layers, and decoded into Hindi tokens — all in seconds.

Transformer Model Architecture

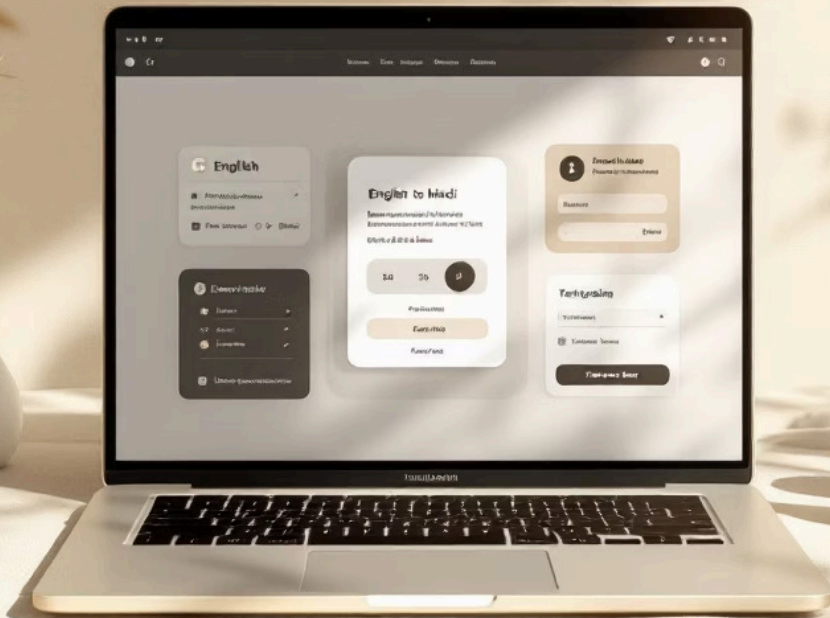


Why Transformers?

Transformers use **self-attention mechanisms** to process entire sequences in parallel — capturing long-range dependencies and context better than traditional sequential models like RNNs.

- Parallel processing for faster inference
- Attention weights capture context
- Pre-trained on massive multilingual data

Live Output



Input (English)

"Hello, how are you today?"

Processing

Tokenized → Encoded →
Translated via transformer
model in real time

Output (Hindi)

"नमस्ते, आज आप कैसे हैं?"

The translation is displayed **instantly** on the web interface, demonstrating the system's real-time capability with no perceptible delay.

System Performance Highlights

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Supported Languages

English, Hindi, French, Spanish, and German

<5s

Response Time

Near-instant real-time translation output

1

Click Interface - Simple UI

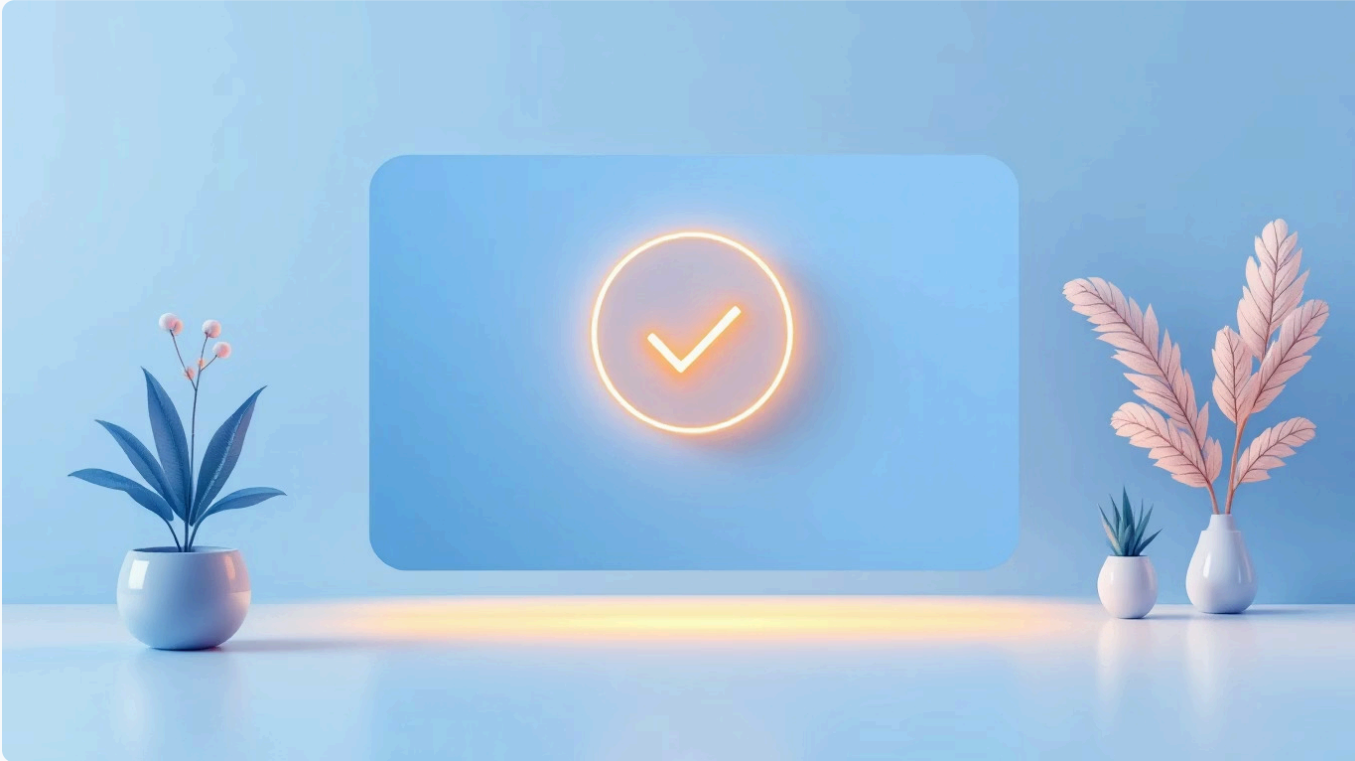
Simple, single-input web form for all users

24/7

Availability

Available through Hugging Face Spaces deployment

Conclusion



What We Built

This project successfully demonstrates the power of **NLP and transformer models** for practical language translation. The system effectively translates English text to Hindi and is deployed for real-world usage.

- ❏ The project validates that pre-trained transformer models can be efficiently used and deployed for low-resource language pairs like English-Hindi.

Future Scope



Regional Language Support

Extend the system to support additional Indian regional languages.



Speech-to-Text Input

Integrate voice recognition for speech-based translation input.



Translation History

Allow users to save and access previous translations.



Mobile Application

Develop a cross-platform mobile app for accessible, on-the-go translation.

